

1.2 Statement of the Problem

“How can Anthropic API-assisted business insight summaries help the owner understand revenue concentration patterns and possible business risks?”

1.3 Objectives of the Study

“Integrate Anthropic API-assisted business insight summaries that explain computed revenue analytics, including top contributors, concentration patterns, and possible business risks for owner review.”

1.6 Definition of Terms, Artificial Intelligence (AI)

“The use of an approved AI service, specifically the Anthropic API, to generate natural-language business insight summaries from computed revenue analytics.”

2.2 Related Technologies, Artificial Intelligence Integration

“Artificial intelligence integration in RevInsight AI is implemented through the Anthropic API, which will generate natural-language insights from summarized revenue analytics. The AI component does not replace the analytics engine and does not require the development of a new machine learning model. Instead, it explains computed results in a way that is easier for the business owner to understand. The Anthropic API supports text generation from prompts, which makes it suitable for turning computed revenue summaries into short business insight explanations. The AI feature receives summarized information such as top revenue sources, percentage contributions, sales trends, and concentration indicators. It then produces observations, decision-support notes, and possible areas for attention. AI-assisted decision support and visualization studies show that clear visual outputs and AI-supported explanations can help users interpret complex information for decision-making. For RevInsight AI, the AI integration supports practical decision-making by explaining whether revenue is concentrated in a few products or services. The generated output will be treated as decision support only, while the owner remains responsible for reviewing and applying the insights.”

2.3 Conceptual Framework

“In the output stage, the system presents the processed information through dashboards, charts, reports, and Anthropic API-assisted business insight summaries.”